

SQL Assignment – Window Functions and CTEs

This assignment focuses on Common Table Expressions (CTEs) and SQL Window Functions. The exercises cover commonly used analytical SQL concepts including: RANK, DENSE_RANK, ROW_NUMBER, PARTITION BY, LAG, LEAD, NTILE, running totals, moving averages, and recursive CTEs.

Database Schema

The dataset models a retail sales company with employees, customers, products, and orders.

employees				
employee_id	employee_name	department	salary	hire_date
1	Alice Johnson	Sales	70000	2020-01-15
2	Bob Smith	Sales	65000	2021-03-20
3	Charlie Brown	IT	90000	2019-07-01
4	Diana Prince	IT	95000	2018-11-11

orders				
order_id	customer_id	employee_id	order_date	total_amount
101	1	1	2024-01-10	500
102	2	2	2024-01-11	700
103	1	1	2024-01-15	1200
104	3	3	2024-01-18	300

Assignment Questions

1. Use ROW_NUMBER() to assign a row number to employees ordered by salary descending.
2. Use RANK() to rank employees by salary.
3. Use DENSE_RANK() to rank employees by salary.
4. Find the top 3 highest-paid employees using a window function.
5. Rank employees within each department using PARTITION BY.
6. Display the highest salary in each department using a window function.
7. Calculate the running total of order amounts ordered by order_date.
8. Calculate the cumulative sales amount for each employee.
9. Use LAG() to show the previous order amount for each customer.
10. Use LEAD() to show the next order amount for each customer.
11. Find the difference between the current order amount and previous order amount.

12. Calculate a moving average of the last 3 orders.
13. Use NTILE(4) to divide employees into salary quartiles.
14. Find the first order placed by each customer using ROW_NUMBER().
15. Find the latest order placed by each customer.
16. Display employee salaries along with department average salary.
17. Find employees earning above their department average salary.
18. Use SUM() OVER(PARTITION BY department) to calculate department payroll.
19. Find the percentage contribution of each employee salary within their department.
20. Use COUNT() OVER() to show total number of employees alongside each row.
21. Create a CTE to calculate total sales per employee.
22. Use a CTE to find employees whose sales exceed the company average.
23. Create multiple CTEs to calculate customer total spending and rankings.
24. Write a recursive CTE to generate numbers from 1 to 10.
25. Use a recursive CTE to display employee hierarchy data.
26. Create a CTE that filters orders above the average order amount.
27. Use a CTE and window function together to rank customers by total spending.
28. Find the second-highest salary in each department.
29. Display the difference between each employee salary and the department maximum salary.
30. Combine CTEs and window functions to find the top-performing employee in each department based on total sales.

Bonus Challenge:

Create a report showing monthly sales trends using: CTEs Running totals LAG() Percentage growth calculations